W

Waste Heat Recovery (*see* Cogeneration)

Waste Management* (*see also* Pollutants, Chemical)

Waste management is a specialized science in itself and also has its own series of handbooks. However, if waste contaminates a site, it is possible that a process engineer may be held liable for the pollution, depending on whether mishaps in his or her plant's process caused it. For that reason, a definition of contaminated site classifications and a user's guide to classification of contaminated sites is included here.

Site Classification Categories

Sites will not be ranked relative to one another. Sites will be classified on their individual characteristics and will be placed into classes (Class 1, 2, 3, or N, according to their priority for action, or Class I for sites that require further information before they can be classified). The classification groupings are as follows:

Class 1 (score 70 to 100): Action required

The available information indicates that action (e.g., further site characterization, risk management, remediation, etc.) is required to address existing concerns. Typically, Class 1 sites show a propensity to high concern for several factors, and measured or observed impacts have been documented.

Class 2 (score 50 to 69.9): Action likely required

The available information indicates that there is high potential for adverse off-site impacts, although the threat to human health and the environment is generally not imminent. There is probably no indication of off-site contamination; however, the potential for this was rated high and therefore some action is likely required.

Class 3 (score 37 to 49.9): Action may be required

The available information indicates that this site is currently not a high concern. However, additional investigation may be carried out to confirm the site classification, and some degree of action may be required.

Class N (score <37): Action not likely required

The available information indicates there is probably no significant environmental impact or human health threats. There is likely no need for action unless new information becomes available indicating greater concerns, in which case the site should be reexamined.

---

* Source: Environment Canada.

TABLE W-1 User's Guide

Category	Evaluation Factor	Scoring Guideline
I. Contaminant(s) characteristics	A. Degree of hazard	
	■ High concern contaminants—high concentration	14
	■ High concern contaminants—low concentration	11
	■ Medium concern contaminants—high concentration	8
	■ Medium concern contaminants—low concentration	5
	■ Low concern contaminants	3
	B. Contaminant quantity (area/volume of site contamination)	
	■ >10 ha or >1000 m^3 or drums of liquid	10
	■ 2 to 10 ha, or 100 to 1000 m^3	6
	■ <2 ha or <100 m^3	2
	C. Physical state of contaminants	
	■ Liquid/gas	9
	■ Sludge	7
	■ Solid	3
	Special considerations	–6 to +6
II. Exposure pathways	A. Groundwater	
	1. Known contamination at or beyond property boundary	
	■ Groundwater significantly exceeds Canadian Drinking Water Guidelines (CDWG) by >2× or known contact of contaminants with groundwater	11
	■ Between 1 and 2× CDWG or probable contact with groundwater	6
	■ Meets Canadian Drinking Water Guidelines	0

Rationale	Method of Evaluation	Sources of Information
In determining the degree of hazard of a waste, it is recognized that a listed hazardous waste is generally of greater concern than a liquid or solid industrial waste. These are in turn of greater concern than other solid wastes. Municipal and organic wastes are considered medium concern contaminants due to their putrescible nature (production of methane and other landfill gases). Household wastes may contain hazardous materials (e.g., batteries, medical wastes, paints, etc).	Determine the level of hazard according to the following table of typical contaminants and definition of high concentrations: *High concern contaminants* ■ Materials defined as dangerous goods in the Transport of Dangerous Goods Act and Regulations ■ Materials identified by Province as hazardous waste (pesticides, herbicides, paint sludge, acid and alkaline solutions, solvents, etc.) ■ Materials regulated by the Canadian Environmental Protection Act (e.g., PCBs) ■ Institutional waste (lab, schools hospitals, etc.) ■ Pathological wastes and animal carcasses ■ Radioactive wastes *Medium concern contaminants* ■ Liquid waste not referred to in above, petroleum products septic tank pumpings, agricultural and chemical containers ■ Food processing wastes ■ Nonhazardous incinerator residues ■ Municipal solid (household) wastes ■ Organic and vegetable wastes ■ Mining residues *Low concern contaminants* ■ Industrial and commercial solid wastes, (e.g., construction materials such as wood, metal, hay, sand/silt piles, etc.) ■ Other nearly inert wastes (e.g., foundry sands) *High concentration of contaminants* ■ Contaminant concentrations in soil, groundwater, or surface water exceed Canadian Environmental Quality Criteria for Contaminated Sites (>2× commercial/ industrial level); or material that was deposited in highly concentrated form (e.g., >5000 ppm)	Transport of Dangerous Goods Act; Provincial/ Territorial Hazardous Wastes lists; regulations under Canadian Environmental Protection Act; Canadian Environmental Quality Criteria for Contaminated Sites; etc.
Little information is known about the quantity of wastes at abandoned sites in Canada. Therefore, waste interpreted from area or quantity information.	Measure or estimate the area or quantity of potential contamination. *Note:* Any number of drums abandoned or disposed is considered a high concern.	
Contaminants in liquid form are more mobile in the ground and water than solids. However, certain water-soluble solid wastes are more mobile than viscous liquids, and these should be evaluated on a case-by-case basis.	Determine the state of the contaminant when it was disposed or deposited. Technical judgment.	
The legislative basis for most jurisdictions is to prevent off-site migration of contamination.	Review chemical data and evaluate groundwater quality. If contamination at or beyond the property boundary exceeds Canadian Drinking Water Guidelines (CDWG) or applicable provincial/territorial guidelines or policies, or if contaminants are known to be in contact with groundwater, then evaluate the site as high.	Canadian Water Quality Guidelines; Provincial/ Territorial Water Quality Guidelines or policies: Guidelines for Canadian Drinking Water Quality.

TABLE W-1 User's Guide (*Continued*)

Category	Evaluation Factor	Scoring Guideline
	2. Potential for groundwater contamination	
	(a) Engineered subsurface containment	
	■ No containment	4
	■ Partial containment	2
	■ Full containment	0
	(b) Thickness of confining layer over aquifer(s) of concern	
	■ 3 m or less	1.5
	■ 3 to 10 m	1
	■ >10 m	0
	(c) Hydraulic conductivity of the confining layer	
	■ $>10^{-4}$ cm/s	1.5
	■ 10^{-4} to 10^{-6} cm/s	1
	■ $<10^{-6}$ cm/s	0.5
	(d) Annual rainfall	
	■ >1000 mm	1
	■ 600 mm	0.6
	■ 400 mm	0.4
	■ 200 mm	0.2
	(e) Hydraulic conductivity of aquifer(s) of concern	
	■ $>10^{-2}$ cm/s	3
	■ 10^{-2}–10^{-4} cm/s	1.5
	■ $<10^{-4}$ cm/s	0.5
	3. Special considerations	–4 to +4
B. Surface water		
	1. Observed or measured contamination of water/effluent discharged from site	
	■ Known or strongly suspected to exceed Canadian Water Quality Guidelines (CWQE) by >2×	11
	■ Known or strongly suspected to be between 1 – 2 × CWQE	6
	■ Meets Canadian Water Quality Guidelines	0
	2. Potential for surface water contamination	
	(a) Surface containment	
	■ No containment	5
	■ Partial containment	3
	■ Full containment	0.5
	(b) Distance to perennial surface water	
	■ 0 to <100 m	3
	■ 100 to 300 m	2
	■ >300 m	0.5

Rationale	Method of Evaluation	Sources of Information
Well contained sites have minimal potential for pollution. Potential for pollution decreases with increasing containment.	Review the existing engineered systems and relate these structures to hydrogeology of the site and determine if full containment is achieved. Full containment is defined as an engineered system, monitored as being effective, which provides for the capture and treatment of contaminants. If there is no system, this factor is evaluated high. If there is less than full containment or if uncertain then evaluate as medium. Typical engineered systems include leachate collection systems and low permeability liners.	
The thickness of a confining layer (e.g., clay, shale, etc.) between contaminants and any aquifers of concern will affect the attenuation of contaminants and hence the quantity and quality of contaminants reaching the aquifers.	Measure or estimate thickness of any confining layer (e.g., clay, shale, etc.) over all aquifers of concern from existing well records or from a general knowledge of local conditions. If possible, an estimate of the continuity of the confining layer should be made from borehole well record information. Note: an aquifer is defined as a geologic material that will yield groundwater in usable quantities.	Historical geologic maps, well records, government hydrogeologist or local consultants.
The rate at which contaminants migrate through the confining layer will affect attenuation and the contaminant loading to the aquifers.	Determine the nature of geologic materials and estimate hydraulic conductivity from published material. Clays, granite, shales should be scored low. Silts etc. should be scored medium. Sand, gravel, and limestone should be scored high.	Freeze and Cherry, 1979, and other groundwater texts.
The quantity of rainfall affects the quantity of leachate produced. Higher leachate quantities have a higher impact on the environment	Refer to Environment Canada rainfall records for relevant areas. Use 30-year average rainfall for evaluation purposes. Divide rainfall by 1000 and round to nearest tenth (e.g., 667 mm = 0.7 score).	Hydrological Atlas of Canada (Fisheries and Environment Canada, 1978).
Aquifers with high hydraulic conductivity can transport contaminants at high velocity over great distances, e.g., solution limestones, highly fractured rocks or gravel deposits.	Determine the nature of geologic materials and estimate hydraulic conductivity of all aquifers of concern from published material.` Technical judgment.	Freeze and Cherry, 1979.
The legislative basis in all jurisdictions is not to contaminate surface water beyond established limits.	Collect all available information on quality of surface water near to site. Evaluate available data against Canadian Water Quality Guidelines (select appropriate guidelines based on local water use, e.g., recreational, irrigation, freshwater aquatic life, etc.) and relevant provincial/ territorial water quality objectives.	CCME Canadian Water Quality Guidelines; Relevant provincial/ territorial and federal legislation and regulations.
The level and type of engineered containment will affect the potential for contaminants to be released to surface water.	Review the existing engineered systems and relate these structures to site conditions and proximity to surface water and determine if full containment is achieved; e.g., evaluate low if there is full containment such as capping, berms, dikes; evaluate medium if there is partial containment such as natural barriers, trees, ditches, sedimentation ponds; evaluate high if there are no intervening barriers between the site and nearby surface water.	Site inspection reports, air photos, etc.
The distance to surface water will affect the probability of contaminants reaching the watercourse. The Ontario Ministry of the Environment has established a classification for immediate impact zone at 50 m. For conservatism, this zone has been broadened to 100 m.	Review available mapping and survey data to determine distance to nearest surface water bodies.	

TABLE W-1 User's Guide (*Continued*)

Category	Evaluation Factor	Scoring Guideline
	(c) Topography	
	■ Contaminants above ground level and slope is steep	1.5
	■ Contaminants at or below ground level and slope is steep	1.2
	■ Contaminants above ground level and slope is flat	0.8
	■ Contaminants at or below ground level and slope is flat	0
	(d) Run-off potential	
	■ >1000 mm rainfall and low permeability surface material	1
	■ 500 to 1000 mm rainfall and moderately permeable surface material	0.6
	■ <500 mm rainfall and highly permeable surface material	0.2
	(e) Flood potential	
	■ 1 in 2 years	0.5
	■ 1 in 10 years	0.3
	■ 1 in 50 years	0.1
	3. Special considerations	–4 to +4
	C. Direct contact	
	1. Known contamination of media off-site	
	■ Known contamination of soil, sediment, or air off-site due to contact with contaminated soil, dust, air, etc. (vector transported should also be considered).	11
	■ Strongly suspected contamination of media off-site	6
	■ No contamination of media off-site	0
	2. Potential for direct human and/or animal contact	
	(a) Airborne emissions (gases, vapors, ducts, etc.)	
	■ Known or suspected airborne emissions impacting on neighboring properties	5
	■ Airborne emissions generally restricted to site	3
	■ No airborne emissions	0
	(b) Accessibility of site (ability to contact materials)	
	■ Limited or no barriers to prevent site access; contaminants not covered	4
	■ Moderate accessibility or intervening barriers; contaminants are covered	3
	■ Controlled access or remote location and contaminants are covered	0
	(c) Hazardous soil gas migration	
	■ Contaminants are putrescible and soil permeability is high	2
	■ Soil contaminants are putrescible but soil permeability is low and/or groundwater is <2 m from surface	1
	■ No putrescible contaminants at the site.	0
	3. Special considerations	–4 to +4
III. Receptors	A. Human and animal uses	
	1. Known adverse impact on humans or domestic animals as a result of the contaminated site	
	■ Known adverse effect on humans or domestic animals	18
	■ Strongly suspected adverse effect on humans or domestic animals	15

Rationale	Method of Evaluation	Sources of Information
Water can run off (and therefore potentially contaminate surface water) with greater ease from elevated sites on slopes.	Review engineering documents on the topography of the site and the slope of surrounding terrain. ■ Steep slope = >50% ■ Flat slope = <5% *Note:* Type of fill placement (e.g., trench, above ground, etc.)	
Run-off transports contaminants into water bodies. Water run-off is a function of precipitation and the rate of infiltration (less permeable soils will allow greater turn-off).	Refer to Environment Canada precipitation records for relevant areas. Use 30-year average precipitation for evaluation purposes.	Hydrological Atlas of Canada (Fisheries and Environment Canada, 1978).
The potential for large quantities and concentrations of contaminants to be released to surface water courses over a short period of time will be affected by the flood potential of a water course near the site.	Review published data such as flood plain mapping or flood potential (e.g., spring or mountain run-off) and Conservation Authority records to evaluate flood potential of nearby water course both up and down gradient. Rate zero if site not in flood plain. Technical judgment.	Established flood plain guidelines/ maps; provincial/ territorial soil survey maps.
Known or measured contamination off-site is an important consideration for determining impact of contaminants.	Record known or measured contamination of soil, sediment, or air on or off-site. Note any presence of soil gas, such as methane, associated with site.	
If air emissions are evident off-site, there is a great hazard for direct contamination of neighboring biota and/or resources.	Review available site information to determine if there have been complaints off-site (due to vapors, gas, dust, etc.). Reports for these problems are not likely available for most abandoned sites. Review regulatory site inspection reports. If airborne emissions are known to be impacting neighboring properties and possibly endangering the public, some immediate action (including characterization of emissions) should be initiated to curtail hazardous emissions or otherwise reduce or eliminate exposure.	Site inspection reports, etc.
The greater the accessibility to a site and to contaminants, the greater the chance for contamination of human and animal life by direct contact.	Review location and engineering of the site and determine if there are intervening barriers between the site and humans or animals. A low rating should be assigned to a (covered) site surrounded by a locked chain link fence or in a remote location, whereas a high score should be assigned to a site that has no cover, fence, natural barriers, or buffer.	
Methane gas migration has been known to cause explosions adjacent to abandoned landfills.	Consider presence of organic material on site, the depth to water table, soil hydraulic conductivity, vegetative stress, odors, etc. Technical judgment.	
Contamination from a site that causes a measurable impact on humans is a great concern.	Review and evaluate reports of impact(s) of site contamination (e.g., increased heavy metal levels measured in blood of nearby residents as a result of site contamination). Any site assigned 15 or more points for this factor should automatically be classified as Class 1. An adverse effect is considered to be any one or more of the following: (i) impairment of the quality of the natural environment for any use that can be made of it, (ii) injury or damage to property or to plant or animal life, (iii) harm or material discomfort to any person, (iv) impairment of the safety of any person, (v) rendering any property or plant or animal life unfit for use by humans, (vi) loss of enjoyment of normal use of property, and (vii) interference with the normal conduct of business (from Ontario Environmental Protection Act, 1980).	

TABLE W-1 User's Guide (*Continued*)

Category	Evaluation Factor	Scoring Guideline
	2. Potential for impact on humans or animals	
	(a) Drinking water supply	
	(i) Known impact on drinking water supply; drinking water supply is known to be adversely affected as a result of site contamination	
	■ Known contamination of drinking water supply to levels above CDWG	9
	■ Strongly suspected contamination of drinking water supply	7
	■ Drinking water supply is known not to be contaminated	0
	(ii) Potential for impact on drinking water supply; proximity to drinking water supply	
	■ 0 to <100 m	6
	■ 100 to <300 m	5
	■ 300 m to <1 km	4
	■ 1 to 5 km	3
	"Availability" of alternate drinking water supply	
	■ Alternate drinking water supply is not available	3
	■ Alternate drinking water supply would be difficult to obtain	2
	■ Alternate drinking water supply available	0.5
	b) Other water resources	
	(i) Known impact on used water resource; water resource (used for recreational purposes, commercial food preparation, livestock watering, irrigation, or other food chain uses) is known to be adversely affected as a result of site contamination.	
	■ Water resource is known to be contaminated above CWQG	4
	■ Water resource is strongly suspected to be contaminated above CWQG	3
	■ Water resource is known not to be contaminated	0
	(ii) Potential for impact on water resources; proximity to water resources used for activities listed above	
	■ 0 to <100 m	2
	■ 100 to <300 m	1.5
	■ 300 m to <1 km	1
	■ 1 to 5 km	0.5
	Use of water resources—if multiple uses, give highest score (use following table)	0.2–2

Water Use	Frequency of Use	
	Frequent	Occasional
Recreational (swimming, fishing)	2	1
Commercial food preparation	1.5	0.8
Livestock watering	1	0.5
Irrigation	1	0.5
Other domestic or food chain uses	0.5	0.3
Not currently used but likely future use	0.5	0.2

	(c) Direct human exposure	
	(i) Known contamination of land used by humans	
	■ Known contamination of land used for agricultural or residential/parkland/school purposes above AG or R/P EQC values	5

Rationale	Method of Evaluation	Sources of Information
Water used for drinking should be protected against contamination from any site.	Review available site data (inspection reports, assessment documentation) to determine if drinking water (groundwater, surface water, private, commercial, or municipal supply) is known or suspected to be contaminated above Guidelines for Canadian Drinking Water Quality or applicable provincial/territorial guidelines or policies. If drinking water supply is known to be contaminated above these guidelines, some immediate action (e.g., provision of alternate drinking water supply) should be initiated to reduce or eliminate exposure.	Guidelines for Canadian Drinking Water Quality; other drinking water guidelines developed by recognized agencies (e.g., other Health and Welfare Canada guidelines, U.S. EPA, etc.).
The nearer a drinking water well is to a contaminant source, the greater the potential for contamination. Well water used for irrigation/agricultural purposes should also be included as it may be used for human consumption.	Review provincial/territorial base mapping or air photos and measure the distance to the nearest resident or drinking water supply. Judge whether the water is being used as a drinking water source. Commonly rural areas use groundwater for drinking purposes. For urban sites, contact the local Public Utilities Commission to determine water source and location.	
This factor takes into account the availability of replacement water supplies, and is used in the technical sense as a factor to indicate the degree of urgency, not as a sociopolitical consideration.	Determine availability of alternate drinking water supply or distance to alternate source.	
The water used for these purposes (groundwater or surface water) should be protected against contamination.	Review documentation for reported or suspected contamination of water used for recreation or food chain uses, and refer to Canadian Water Quality Guidelines or other relevant guidelines (select appropriate guidelines based on local water used) to determine if supply is considered contaminated.	CCME Canadian Water Quality Guidelines; provincial/ territorial water quality guidelines and objectives; etc.
The nearer a water resource is to a site, the greater the risk of contamination.	Determine distance from the site to the nearest recreational or food chain used water resource.	
Potential for impact due to use of water resource is related to the type and frequency of use. Human uses are of the highest cencern.	Assess water users adjacent to the site from maps and directories.	
Hazards associated with soil contamination are directly related to land use.	Review zoning and land use maps for lands adjacent the site. Evaluate levels of soil contamination against Canadian Environmental Quality Criteria (EQC) for Contaminated Site (AG = agricultural level; R/P = residential/parkland	CCME Canadian Environmental Quality Criteria for Contaminated Sites.

TABLE W-1 User's Guide (*Continued*)

Category	Evaluation Factor/Guideline	Scoring Rationale
	■ Known contamination of land used for commercial or industrial purposes above C/I EQC values	3.5
	■ Land is known not to be contaminated	0
	(ii) Potential human exposure through land use	
	■ Use of land at and surrounding site (use following table; give highest score to worst case scenario)	0.5–5

	Distance from Site		
Lane use (current or future)	0–300 m	300 m–1 km	1–5 km
Residential	5	4.5	3
Agricultural	5	4	2.5
Parkland/school	4	3	1.5
Commercial/industrial	3	1	0.5

3. Special considerations	–5 to +5
B. Environment	
1. Known adverse impact on a sensitive environment as a result of the contaminated site	
■ Known adverse impact on sensitive environment	16
■ Evidence of stress on aquatic species or vegetative stress on trees, crops, or plant life located on properties neighboring the site	14
■ Strongly suspected adverse impact on sensitive environment	12
2. Potential for impact on sensitive environments	
(a) Distance from site to nearest sensitive environment (e.g., sensitive aquatic environment, nature preserve, habitat for endangered species, sensitive forest reserves, national parks or forests, etc.)	
■ 0 to <500 m	10
■ 500 m to <2 km	6
■ 2 to <5 km	2
■ 5 to 10 km	0.5
(b) Groundwater—distance to important or susceptible groundwater resource(s)	
■ 0 to <500 m	6
■ 500 m to <2 km	4
■ 2 to <5 km	2
■ 5 to 10 km	1
3. Special considerations	–5 to +5

Method of Evaluation	Information	Sources of
	level; C/I = commercial/industrial level). If soil is known to be contaminated above these levels and possibly endangering public health, some immediate action (e.g., fencing the area, limiting public access, etc.) should be initiated to reduce or eliminate the exposure.	
Hazards associated with soil contamination are directly related to land use and distance of the used land from the site. Residential and agricultural land uses are of highest concern because humans are situated at these locations for longer periods.	Review zoning and land use maps over the distances indicated. If the proposed future land use is more "sensitive" than the current land use, evaluate this factor assuming the proposed future use is in place (indicate in the worksheet that future land use is the consideration). Agricultural land use is defined as uses of land where the activities are related to the productive capability of the land or facility (e.g., greenhouse) and are agricultural in nature, or activities related to the feeding and housing of animals as livestock. Residential/Parkland land uses are defined as uses of land on which dwelling on a permanent, temporary, or seasonal basis is the activity (residential), as well as uses on which the activities are recreational in nature and require the natural or human designed capability of the land to sustain that activity (parkland). Commercial/Industrial land uses are defined as land on which the activities are related to the buying, selling, or trading of merchandise or services (commercial), as well as land uses that are related to the production, manufacture, or storage of materials (industrial). Technical judgment.	
The environment should be to protected against site contamination. Evidence of impact(s) shows that protection is lacking.	Review records for evidence of vegetative stress or impairment of any nearby sensitive environments. A sensitive environment is defined as a sensitive aquatic environment, nature preserve, habitat for endangered species, sensitive forest reserves, national parks or forests, etc. An adverse effect is considered to be any one or more of the following: (i) impairment of the quality of the natural environment for any use that can be made of it, (ii) injury or damage to property or to plant or animal life, (iii) harm or material discomfort to any person, (iv) impairment of the safety of any person, (v) rendering any property or plant or animal life unfit for use by humans, (vi) loss of enjoyment of normal use of property, and (vii) interference with the normal conduct of business (from Ontario Environmental Protection Act, 1980).	
It is considered that within approximately 1 km of the site there is immediate concern for contamination. Therefore, an environmentally sensitive area located within this area of the site will be subject to concern. It is also generally considered that any sensitive area located greater than 10 km from the site will not be impacted.	Review Conservation Authority mapping and literature. Also review Ministry of Natural Resources records and Federal Land Capability maps. Identify provincial/territorial and federal designated environmentally sensitive areas.	Relevant provincial/ territorial and federal maps of sensitive environments.
The closer a site is to a discharge or recharge area, the greater the potential for contamination of a groundwater or surface water resource.	Review groundwater contour maps, if available, and other available reports. Otherwise use established hydrogeologic principles. Technical judgment.	Local groundwater maps, etc.

Class I (estimated score ≥15): Insufficient information

There is insufficient information to classify the site. In this event, additional information is required to address data gaps.

Within each priority category, further refinement of the relative classification of sites may be necessary. The National Classification System is a screening tool only. Firm conclusions about the need for remedial action will still depend on a number of factors (including planned long-term use or redevelopment of the site, application of contaminated site criteria, and relevant/site-specific objectives of the jurisdiction in which the site is located, local issues, availability of technology, remediation costs, etc.). These factors are beyond the scope of this section, however, and are not meant to be addressed.

See Table W-1.

Reference and Additional Reading

1. Soares, C. M., *Environmental Technology and Economics: Sustainable Development in Industry*, Butterworth-Heinemann, 1999.

Welding (*see* Metallurgy)